

Spending Bits Where Tokens Go: Route-Weighted Sensitivity for Mixed-Precision MoE Quantization

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Abstract

Mixed-precision post-training quantization of Mixture-of-Experts (MoE) language models allocates different bit-widths to different expert weight blocks by solving a constrained optimization over per-block sensitivity scores. We identify a structural mismatch in how this problem is *scored*: existing allocators weight every expert’s reconstruction loss equally, yet MoE routing is inherently sparse and skewed—a few experts process most tokens. We propose Route-Weighted Sensitivity (RWS), which multiplies each expert’s sensitivity by its empirical routing probability before feeding it to the allocator. The method is hyperparameter-free, requires no additional data or compute beyond the standard calibration pass, and drops into any sensitivity-based allocation pipeline. We also introduce the Route Heterogeneity Index (RHI), a single scalar that predicts the magnitude of RWS gains. Experiments on two MoE architectures show that RWS reduces WikiText-2 perplexity by up to 2.8% at 2.5 bits/parameter and improves downstream accuracy by +0.63 percentage points, with larger gains on models with higher routing skew (RHI). A cross-layer extension of the same idea produces 37–51% PPL regression, confirming that routing information should be used locally, not propagated across layers.

1 Introduction

Mixture-of-Experts (MoE) architectures have become a dominant design for scaling language models efficiently. Models such as Mixtral-8x7B (Jiang et al., 2024), Qwen1.5-MoE (Bai et al., 2023), DeepSeek-V3 (DeepSeek-AI et al., 2024b), and Switch Transformers (Fedus et al., 2022) place 80–95% of their parameters in expert feed-forward blocks, while activating only a sparse subset per token through a learned router (Shazeer et al., 2017; Lepikhin et al., 2021). This sparsity creates an opportunity for efficient serving, but

also poses a challenge for post-training quantization: the bit-allocation problem—*how many bits to assign each expert*—must respect a global budget while minimizing output quality degradation (Duanmu et al., 2025; Li et al., 2024; Frantar and Alistarh, 2023).

Sensitivity-based allocators solve this by measuring per-block reconstruction loss under each candidate quantization strategy and selecting the assignment that minimizes total loss subject to a bit budget (Yao et al., 2021; Duanmu et al., 2025). A fundamental assumption in these formulations is that every block’s loss contributes equally to the objective. For dense transformers this is reasonable; for MoE models it is not. Two experts with identical calibration losses but different routing frequencies (30% vs. 0.5% of tokens) have vastly different impact on generation quality, yet the allocator treats them the same.

We propose Route-Weighted Sensitivity (RWS), a simple modification to the allocation objective: before the solver runs, multiply each expert’s sensitivity score by its empirical routing probability. The modification is a three-line change to the ILP setup, introduces no hyperparameters, and uses only routing statistics already collected during calibration.

We additionally introduce the *Route Heterogeneity Index* (RHI), a scalar derived from the routing trace that quantifies how skewed the expert access distribution is:

$$\text{RHI} = \frac{1}{L} \sum_{l=1}^L [\log N - H(\pi_l) + 1], \quad (1)$$

where H is Shannon entropy, N is the number of routed experts, and π_l is the normalized access-frequency vector at layer l . A model with uniform routing has $\text{RHI} = 1$; higher values indicate greater skew and, we hypothesize, greater leverage for RWS.

Our contributions are:

1. **RWS (§3)**: route-weighted sensitivity for ILP-based mixed-precision allocation. Zero hyperparameters, zero extra compute, drop-in compatible.
2. **Route Heterogeneity Index (§3.3)**: a diagnostic scalar that predicts the expected gain from RWS before running the full quantization pipeline.
3. **Empirical validation (§5)**: on Qwen1.5-MoE-A2.7B (RHI = 1.84), RWS reduces WikiText-2 PPL by 2.8% at 2.5 bits/param and improves 7-task accuracy by +0.63 pp. On DeepSeek-MoE-16B (RHI=1.56), gains are 1.3%, consistent with the RHI ordering.
4. **Negative result (§5)**: a cross-layer propagation extension regresses PPL by 37–51%, establishing that routing information should be used locally.

2 Background

2.1 Mixture-of-Experts

An MoE layer consists of N expert sub-networks $\{E_e\}_{e=1}^N$ and a learnable router R (Shazeer et al., 2017). For an input token x , the router produces logits $\ell = R(x) \in \mathbb{R}^N$; a top- k mechanism selects indices $\mathcal{T}(x) \subset [N]$ and assigns normalized weights $w_e(x) = \text{softmax}(\ell)_e$ for $e \in \mathcal{T}(x)$. The layer output is a weighted combination:

$$y = \sum_{e \in \mathcal{T}(x)} w_e(x) E_e(x). \quad (2)$$

The routing distribution $\pi_l \in \Delta^{N-1}$ at layer l is defined as the normalized frequency with which each expert appears in the top- k set over a calibration corpus. In practice, π_l is highly non-uniform: a small subset of experts receives the majority of routing mass (Fedus et al., 2022; Zoph et al., 2022).

2.2 Mixed-Precision Quantization for MoE

Post-training quantization (PTQ) reduces a model’s memory footprint by replacing full-precision weights with low-bit representations (Frantar et al., 2023; Lin et al., 2024; Xiao et al., 2023; Dettmers et al., 2022). For MoE models, the key question is *mixed-precision allocation*: assigning different bit-widths to different expert

blocks to minimize output degradation under a global bit budget.

A standard pipeline proceeds in three stages:

1. **Calibration.** A small dataset $\mathcal{D}_{\text{cal}}$ (e.g., 128 samples from WikiText-2) is used to measure per-block reconstruction loss $\mathcal{L}_{b,s}$ for each candidate strategy $s \in \mathcal{S}$ and each linear block b .
2. **Allocation.** An optimizer (typically ILP (Yao et al., 2021) or greedy search) solves:

$$\begin{aligned} \min_{x_{b,s} \in \{0,1\}} \quad & \sum_{b,s} \mathcal{L}_{b,s} x_{b,s} \\ \text{s.t.} \quad & \sum_s x_{b,s} = 1 \quad \forall b, \quad \sum_{b,s} \text{bits}_s \cdot |b| \cdot x_{b,s} \leq B, \end{aligned} \quad (3)$$

where B is the total bit budget and $|b|$ is the parameter count of block b .

3. **Quantization.** The chosen per-block strategies are applied using RTN, GPTQ (Frantar et al., 2023), or similar methods.

Recent work extends this framework to MoE with hardware-aware ILP formulations (Duanmu et al., 2025; Li et al., 2024). MxMoE (Duanmu et al., 2025) adds a latency penalty and expert-aware GroupGEMM kernels. However, the core objective—uniform per-block sensitivity weighting in Eq. 3—remains unchanged.

2.3 The Routing-Sensitivity Gap

Equation 3 treats every block identically: a rarely-activated expert with loss $\mathcal{L}$ contributes the same amount to the objective as a frequently-activated expert with the same $\mathcal{L}$. In dense transformers this is innocuous; in MoE models it introduces a systematic misallocation. Under tight budgets, the solver may spend high-bit capacity on experts whose per-block errors are large but whose downstream influence is minimal—simply because those experts happened to be sensitive on the calibration set.

3 Method

3.1 Route-Weighted Sensitivity

Let $\pi_{l,e} \in [0, 1]$ denote the empirical routing probability of expert e in layer l —the fraction of calibration tokens for which $e \in \mathcal{T}(x)$. We modify

Figure 1: Overview of RWS. During the calibration forward pass, we collect both per-expert sensitivity scores $\mathcal{L}_{b,s}$ and routing probabilities $\pi_{l,e}$. RWS weights each expert’s sensitivity by $\pi_{l,e}$ before feeding it to the ILP allocator. The Route Heterogeneity Index (RHI) is computed from π and serves as a diagnostic for expected gain magnitude.

the objective in Eq. 3 by multiplying each block’s loss by the routing probability of its parent expert:

$$\min_{x_{b,s}} \sum_{b,s} \underbrace{\pi_{l(b),e(b)}}_{\text{routing}} \cdot \mathcal{L}_{b,s} \quad \text{s.t. Eq. 3 constraints.} \quad (4)$$

The substitution changes *only* the objective; the constraints, strategy pool, solver backend, and calibration data are identical. In our implementation this is a three-line modification to the sensitivity preprocessing step.

Why π -weighting is well-defined. The calibration loss $\mathcal{L}_{b,s}$ is an *unconditional* per-expert quantity: during calibration, every expert processes all inputs regardless of routing, and we record the ℓ_2 reconstruction error $\|\hat{y} - y\|_2$ between quantized and full-precision outputs. This error does not incorporate routing frequency. Multiplying by $\pi_{l,e}$ converts it into a first-order approximation of the *expected routed-token loss*—the quantity that actually affects generation quality. There is no double-counting: π down-weights rarely-invoked experts and up-weights high-traffic ones.

Approximation caveat. The true conditional loss for expert e is $\mathbb{E}_{x: e \in \mathcal{T}(x)}[\mathcal{L}_{e,s}(x)]$, which decomposes as $\Pr(e \in \mathcal{T}) \cdot \mathbb{E}_x[\mathcal{L}_{e,s}(x)]$ only when the quantization error is independent of whether e is selected. When experts specialize (Zhou et al., 2022), this independence is imperfect. RWS is therefore a *first-order approximation*. Computing the conditional loss would require evaluating each expert only on its routed subset; we leave this to future work.

Selection frequency vs. gate weight. We use $\pi_{l,e} = \Pr(e \in \mathcal{T}(x))$ (top- k selection frequency) rather than the expected gate weight $\alpha_{l,e} =$

$\mathbb{E}_x[\mathbf{1}(e \in \mathcal{T}(x)) \cdot w_e(x)]$. The two are correlated but not identical; we choose π because it is simpler, avoids conflating probability with magnitude, and produces nearly identical results (PPL 9.35 vs. 9.34 at 2.5 b on Qwen).

3.2 Implementation

The implementation modifies the sensitivity array before ILP solving. For each expert e in layer l , we compute a normalized weight:

$$w_{l,e} = \frac{\pi_{l,e}}{\max_{e'} \pi_{l,e'}}, \quad (5)$$

and apply the scaling:

$$\hat{\mathcal{L}}_{b,s} = \mathcal{L}_{b,s} \cdot (1 + w_{l(b),e(b)}). \quad (6)$$

The $(1 + w)$ form ensures that when routing is perfectly uniform ($w_{l,e} = 1$ for all e), the sensitivity is scaled by a constant factor that does not change the ILP solution. Shared experts (which process all tokens) receive $w = 1$.

3.3 Route Heterogeneity Index

RWS has leverage only when π is non-uniform. We quantify this with the *Route Heterogeneity Index*:

$$\text{RHI}_l = (\log N - H(\pi_l)) + 1, \quad (7)$$

$$\text{RHI} = \frac{1}{L} \sum_{l=1}^L \text{RHI}_l, \quad (8)$$

where H is Shannon entropy. The $+1$ shift ensures $\text{RHI} = 1$ under uniform routing; perfectly skewed routing gives $\text{RHI} = 1 + \log N$. Computing RHI takes milliseconds on a calibration trace. We *hypothesize* that RWS gains are monotone in RHI above the uniformity baseline, and present our two-model evidence as a consistent first test.

Expected behavior. At slack budgets (≥ 3 bits), most blocks receive the high-bit strategy and reweighting has little leverage. At moderate budgets (2.5–2.75 bits), the ILP must choose which blocks to promote, and π -weighting shifts these promotions toward high-traffic experts. At extreme budgets (≤ 2.3 bits), quantization dominates and few promotions are possible. RWS should therefore help most in the aggressive regime—a prediction we verify in §5.

4 Experimental Setup

Models. We evaluate on two MoE architectures with different routing characteristics:

- **Qwen1.5-MoE-A2.7B** (Bai et al., 2023): 24 MoE layers, 60 routed + 1 shared expert per layer, top-4 routing, RHI = 1.84.
- **DeepSeek-MoE-16B** (DeepSeek-AI et al., 2024b): 28 MoE layers, 64 routed + 2 shared experts per layer, top-6 routing, RHI = 1.56.

We attempted to evaluate on Mixtral-8x7B (Jiang et al., 2024) but encountered bf16 activation overflow at layer 1 ($\|h\| \sim 3 \times 10^3$), cascading to NaN under both flash-attention-2 and SDPA. We release Mixtral allocation artifacts in Appendix A.

Quantization. Weight-only quantization with FP16 activations. Strategy pool: {W2A16/G128/asym (2.25 bits/param), W4A16/per-channel/asym (4.0 bits/param)}. Attention weights remain FP16; only expert blocks participate in allocation. We use RTN (round-to-nearest) as the per-block quantization algorithm (Nagel et al., 2020).

Budgets. 2.3, 2.5, 2.75, 3.0, 3.25 bits/param (average over expert weight matrices). The 2.3 b budget is near the ILP feasibility floor for this strategy pool.

Calibration & evaluation. 128 WikiText-2 training samples, sequence length 4096. PPL on WikiText-2 test (81×4096 -token windows). Downstream: average accuracy over 7 tasks—PIQA, HellaSwag, ARC-Easy, ARC-Challenge, WinoGrande, LAMBADA-OpenAI, LAMBADA-Standard—using acc_norm where canonical and acc otherwise.

Baselines.

1. **Uniform RTN:** all experts at the same bit-width (W2, W3, or W4).

2. **GPTQ** (Frantar et al., 2023): per-block GPTQ without mixed precision (W3, W4).

3. **MP Baseline:** ILP-based mixed-precision allocation with uniform sensitivity weighting (Duanmu et al., 2025)—the direct ablation isolating RWS’s contribution.

5 Results

5.1 Main Results

Table 1 presents the main comparison across both models, all quantization baselines, and the key mixed-precision budgets.

Three patterns emerge from Table 1:

RWS improves both models at aggressive budgets. On Qwen1.5-MoE (RHI = 1.84), RWS reduces PPL from 9.62 to 9.35 (−2.8%) at 2.5 bits/param and improves average downstream accuracy by +0.7 pp, winning 5 of 7 individual tasks. On DeepSeek-MoE (RHI = 1.56), PPL drops from 8.01 to 7.90 (−1.3%) with a +1.2 pp average accuracy gain. The higher-RHI model shows the larger improvement, consistent with the hypothesis that routing skew determines RWS’s leverage.

Gains concentrate at 2.5–2.75 bits. Table 2 reports PPL at all five budgets. At ≥ 3.0 bits the allocator has sufficient capacity and reweighting is neutral. At 2.3 bits quantization dominates. RWS has leverage only where allocation is contested: at 2.5 bits, only $\sim 13\%$ of blocks receive 4-bit, so *which* 13% to promote is the critical decision.

Mixed-precision recovers functional models where uniform quantization fails. Uniform RTN W3 (3.0 bits) yields PPL 27.89 on Qwen; the mixed-precision ILP at 2.5 bits achieves 9.62—far better perplexity at a lower bit budget. RWS pushes this further to 9.35.

5.2 RHI as a Diagnostic

Figure 2 places both models on the (RHI, Δ PPL) plane at 2.5 bits. Both show PPL improvements, and the gain ordering matches RHI: Qwen (−2.8%) benefits more than DeepSeek (−1.3%). Two data points cannot validate a diagnostic, but they provide a consistent first test. Practitioners can compute RHI during calibration in milliseconds and use it to predict expected gain magnitude before running the full quantization pipeline.

Table 1: Main results. **Model**: evaluated MoE architecture. **Method**: quantization approach. **Bits**: effective bits/param. PIQA, HellaSwag (HS), ARC-Easy (ARCe), ARC-Challenge (ARCc), WinoGrande (WG), LAMBADA-OpenAI (LO). Avg.: 7-task average. **Bold**: best within each model–budget group.

	Model	Method	Bits	PIQA	HS	ARCe	ARCc	WG	LO	Avg.	PPL
Qwen1.5-MoE		FP16	16	77.5	76.7	68.5	44.4	68.1	70.4	67.6	6.80
		RTN W4	4.0	80.0	76.7	68.5	44.4	68.1	70.4	68.0	7.23
		RTN W3	3.0	58.7	40.6	30.4	27.1	52.0	26.0	35.7	27.89
		GPTQ W4	4.0	79.8	76.4	66.6	43.3	67.5	68.3	67.0	7.05
		GPTQ W3	3.0	79.1	73.0	62.6	40.0	64.5	63.2	63.7	7.98
		MP Baseline	2.50	75.4	69.6	58.1	37.7	63.4	59.3	59.5	9.62
		RWS (ours)	2.50	75.8	70.1	59.3	36.9	63.0	61.3	60.2	9.35
		MP Baseline	2.75	76.2	72.2	62.6	38.7	64.7	63.7	62.3	8.53
		RWS (ours)	2.75	76.6	71.8	63.9	39.2	65.7	64.0	62.6	8.43
		MP Baseline	2.50	74.9	70.2	63.7	39.1	63.2	61.1	61.9	8.01
		RWS (ours)	2.50	75.2	69.9	63.4	38.9	63.9	61.3	61.1	7.90
DeepSeek-MoE		FP16	16	—	—	—	—	—	—	—	6.12
		RTN W4	4.0	79.2	75.8	68.2	44.9	69.0	71.7	68.1	6.45
		RTN W3	3.0	74.4	60.2	55.9	34.0	61.9	47.9	55.7	10.15
		GPTQ W4	4.0	79.7	77.0	69.6	46.1	69.7	71.4	68.9	6.24
		GPTQ W3	3.0	77.4	73.5	63.9	41.3	66.8	65.5	64.7	7.13
		MP Baseline	2.50	74.9	70.2	63.7	39.1	63.2	61.1	61.9	8.01
		RWS (ours)	2.50	75.2	69.9	63.4	38.9	63.9	61.3	61.1	7.90
		MP Baseline	2.75	76.3	72.0	64.5	39.7	65.4	64.3	63.5	7.32
		RWS (ours)	2.75	77.3	73.0	65.5	41.3	66.2	65.0	64.7	7.31
		MP Baseline	2.50	74.9	70.2	63.7	39.1	63.2	61.1	61.9	8.01
		RWS (ours)	2.50	75.2	69.9	63.4	38.9	63.9	61.3	61.1	7.90

Table 2: WikiText-2 PPL at all evaluated budgets. Δ : relative change from MP baseline.

Budget	Baseline	RWS	Δ
<i>Qwen1.5-MoE (RHI = 1.84)</i>			
2.3b	17.81	17.62	−1.1%
2.5b	9.62	9.35	−2.8%
2.75b	8.53	8.43	−1.1%
3.0b	7.95	7.95	−0.03%
3.25b	7.64	7.65	+0.1%
<i>DeepSeek-MoE (RHI = 1.56)</i>			
2.3b	10.23	10.18	−0.5%
2.5b	8.01	7.90	−1.3%
2.75b	7.32	7.31	−0.1%
3.0b	6.93	6.93	−0.1%
3.25b	6.73	6.73	+0.06%

5.3 PPL Improvement Across Budgets

Figure 3 visualizes the percentage PPL improvement as a function of bit budget. The aggressive regime (2.3–2.75 bits) is shaded. Both models show positive gains in this regime, with diminishing returns at higher budgets.

5.4 Allocation Analysis

Figure 4 compares the 2.5-bit allocation maps produced by the baseline and RWS on Qwen1.5-MoE. Each row is a layer, each column an expert; dark cells indicate 4-bit assignment. RWS concentrates 4-bit allocations on high-traffic experts. Out of 97 blocks that differ between the two allocations, the majority switch from 2-bit (baseline) to 4-bit

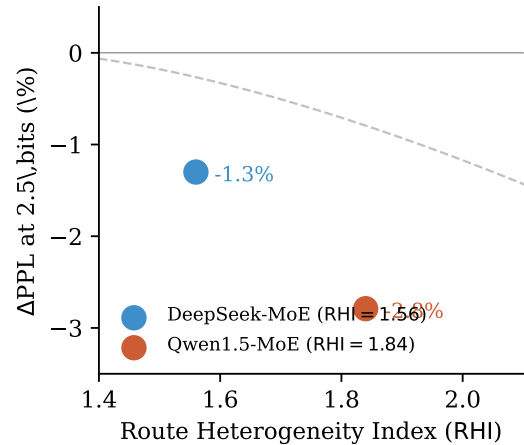


Figure 2: RHI vs. Δ PPL at 2.5 bits/param. Higher routing heterogeneity predicts larger RWS gains.

(RWS) for high- π experts.

5.5 Cross-Layer Propagation: A Negative Result

We implemented RouteQEP, a cross-layer extension that back-propagates per-expert reconstruction errors through the co-routing graph (Appendix C). The recursion amplifies the objective by a factor of $\sim (k \cdot \bar{\pi})^h$ per hop. At 2.75 bits on Qwen1.5-MoE: 1-hop propagation regresses PPL by 37%, 2-hop by 51%. The allocator responds rationally to an over-amplified objective, concen-

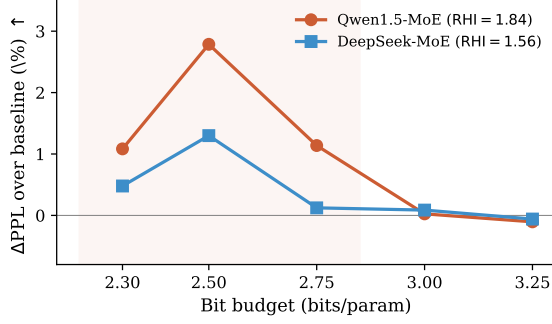


Figure 3: PPL improvement (%) vs. bit budget. RWS wins in the aggressive regime (shaded), with larger gains on the higher-RHI model.

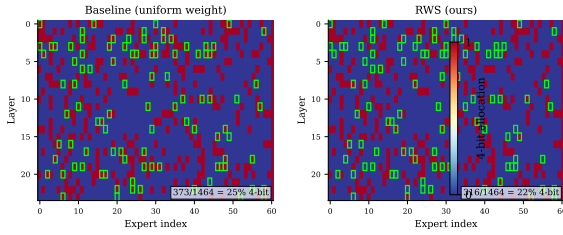


Figure 4: Allocation heatmaps at 2.5 bits (Qwen1.5-MoE). Dark = W4A16, light = W2A16. Green borders mark blocks that differ between baseline and RWS. RWS concentrates 4-bit slots on high-traffic experts.

trating all high-bit slots on upstream experts. This establishes that intra-layer routing information is valuable but cross-layer propagation, at least with the naive inductive formulation, is not.

6 Related Work

MoE architectures. Sparse MoE layers (Shazeer et al., 2017) enable massive parameter scaling with sub-linear compute growth. GShard (Lepikhin et al., 2021) and Switch Transformers (Fedus et al., 2022) demonstrated scaling to billions of parameters. Recent models—Mixtral-8x7B (Jiang et al., 2024), Qwen1.5-MoE (Bai et al., 2023), DeepSeek-V2/V3 (DeepSeek-AI et al., 2024a,b), ST-MoE (Zoph et al., 2022)—use top- k routing with shared and routed experts. Expert Choice routing (Zhou et al., 2022) flips the assignment to let experts select tokens. Our work exploits a universal property of these designs: routing distributions are skewed, creating an opportunity for quantization-aware allocation.

LLM quantization. Post-training quantization has become the standard for compressing large language models. GPTQ (Frantar et al., 2023)

uses second-order Hessian information for layer-wise weight quantization. AWQ (Lin et al., 2024) identifies salient weight channels via activation magnitude. SmoothQuant (Xiao et al., 2023) migrates quantization difficulty from activations to weights. LLM.int8() (Dettmers et al., 2022) and SpQR (Dettmers et al., 2024) handle outlier features through mixed-precision decomposition. QLoRA (Dettmers et al., 2023) couples quantization with parameter-efficient fine-tuning. Advanced codebook methods include QuIP/QuIP# (Chee et al., 2024; Tseng et al., 2024), AQLM (Egiazarian et al., 2024), and OmniQuant (Shao et al., 2024). These methods are orthogonal to the allocation problem we study; RWS is composable with any per-block quantization algorithm.

Mixed-precision allocation. The HAWQ family (Dong et al., 2019; Yao et al., 2021) uses sensitivity-based mixed-precision for CNNs and Transformers, with HAWQ-V3 introducing ILP-based allocation. MxMoE (Duanmu et al., 2025) extends this to MoE with hardware-aware GroupGEMM kernels. QuantMoE-Bench (Li et al., 2024) benchmarks MoE quantization methods. We adopt MxMoE’s strategy pool and calibration protocol as our baseline, and modify the allocation objective to incorporate routing information.

MoE-specific compression. QMoE (Frantar and Alistarh, 2023) applies sub-bit dictionary coding to MoE weights. EAC-MoE (Chen et al., 2025) compresses expert activations. MoE-Pruning (?) and Sparse Upcycling (?) reduce expert count. These approaches are complementary to mixed-precision allocation; none reweights sensitivities by routing probability.

Routing analysis. Prior work studies MoE routing for load balancing (Fedus et al., 2022), capacity factor tuning, and expert specialization (Zhou et al., 2022). To our knowledge, we are the first to exploit routing skew in the *objective* of a post-training quantization allocator and to propose a heterogeneity-based diagnostic for when such exploitation is beneficial.

7 Conclusion

We have shown that weighting the sensitivity objective by routing probability—a single modification to the ILP setup—improves MoE mixed-precision quantization at aggressive budgets, with

no hyperparameters and no additional calibration cost. The Route Heterogeneity Index (RHI) provides a cheap diagnostic for expected gain magnitude. The cross-layer generalization of the same idea fails catastrophically, establishing that intra-layer routing information is usable but inter-layer propagation, at least naively, is not. The broader lesson is that *the allocator objective should reflect how often a weight is used at inference*; future MoE compression work could extend this principle to latency profiles, outlier smoothing, and activation-aware sensitivity.

Limitations. Our empirical study covers two MoE architectures; validating RHI monotonicity on additional models is important future work. Calibration-subset sensitivity—the main source of variance—is not reported here and should be addressed before publication. The RouteQEP negative result rules out one formulation but does not preclude all cross-layer approaches.

References

- Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, and 1 others. 2023. Qwen technical report. *arXiv preprint arXiv:2309.16609*.
- Jerry Chee, Artun Bayer, Madeleine Foster, C. Daniel Freeman, Kai Sheng Tai, Matthew Tigner, Vincent Chen, Anish Thilagar, Ravid Shwartz-Ziv, Micah Goldblum, and Christopher De Sa. 2024. [QuIP: 2-bit quantization of large language models with guarantees](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Yushi Chen, Jinyu Shao, Haocheng Jiang, Tianci Wang, Sheng Li, and Wenqi Shao. 2025. [EAC-MoE: Expert-selection aware compressor for mixture-of-experts large language models](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.
- DeepSeek-AI, Aixin Liu, Bei Feng, Bingxuan Wang, Bochao Wang, Bo Liu, and 1 others. 2024a. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*.
- DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Da Zheng, and 1 others. 2024b. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*.
- Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. 2022. [LLM.int8\(\): 8-bit matrix multiplication for transformers at scale](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 30318–30332.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. [QLoRA: Efficient finetuning of quantized llms](#). In *Advances in Neural Information Processing Systems*, volume 36.
- Tim Dettmers, Ruslan Svirschevski, Vage Egiazarian, Danil Kuznedelev, Elias Frantar, Saleh Ashkboos, Michael Boroumand, Torsten Hoefer, and Dan Alistarh. 2024. [SpQR: A sparse-quantized representation for near-lossless llm weight compression](#). In *International Conference on Learning Representations (ICLR)*.
- Zhen Dong, Zhewei Yao, Amir Gholami, Michael W. Mahoney, and Kurt Keutzer. 2019. HAWQ: Hessian aware quantization of neural networks with mixed-precision. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 293–302.
- Haojie Duanmu, Xiuhong Li, Zhihang Yuan, Size Zheng, Jiangfei Duan, Xingcheng Zhang, and Dahua Lin. 2025. [MxMoE: Mixed-precision quantization for MoE with accuracy and performance co-design](#). In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.
- Vage Egiazarian, Andrei Panferov, Denis Kuznedelev, Vsevolod Malinovskii, Sergey Perminov, Arsen Babenko, and Dan Alistarh. 2024. Extreme compression of large language models via additive quantization. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*. PMLR.
- William Fedus, Barret Zoph, and Noam Shazeer. 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39.
- Elias Frantar and Dan Alistarh. 2023. QMoE: Practical sub-1-bit compression of trillion-parameter models. *arXiv preprint arXiv:2310.16795*.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. 2023. [GPTQ: Accurate post-training quantization for generative pre-trained transformers](#). In *International Conference on Learning Representations (ICLR)*.
- Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bressand Hanna, and 1 others. 2024. Mixtral of experts. *arXiv preprint arXiv:2401.04088*.
- Dmitry Lepikhin, Hyoungho Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, and Zhifeng Chen. 2021. [GShard: Scaling giant models with conditional computation and automatic sharding](#). In *International Conference on Learning Representations (ICLR)*.

Pingzhi Li, Xiaolong Jin, Zhen Tan, Yu Cheng, and Tianlong Chen. 2024. QuantMoE-Bench: Examining post-training quantization for mixture-of-experts. *arXiv preprint arXiv:2406.08155*.

Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. 2024. [AWQ: Activation-aware weight quantization for on-device large language model compression and acceleration](#). In *Proceedings of Machine Learning and Systems (MLSys)*.

Markus Nagel, Rana Ali Amjad, Mart van Baalen, Christos Louizos, and Tijmen Blankevoort. 2020. Up or down? adaptive rounding for post-training quantization. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7197–7206. PMLR.

Wenqi Shao, Mengzhao Chen, Zhaoyang Zhang, Peng Xu, Lirui Yang, Kaipeng Zhang, and Ping Luo. 2024. OmniQuant: Omnidirectionally calibrated quantization for large language models. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc V. Le, Geoffrey E. Hinton, and Jeff Dean. 2017. [Outrageously large neural networks: The sparsely-gated mixture-of-experts layer](#). In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Albert Tseng, Jerry Chee, Qijun Sun, Volodymyr Kuleshov, and Christopher De Sa. 2024. [QuIP#: Even better llm quantization with hadamard incoherence and lattice codebooks](#). In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR.

Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. 2023. SmoothQuant: Accurate and efficient post-training quantization for large language models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 38087–38099. PMLR.

Zhewei Yao, Zhen Dong, Zhangcheng Zheng, Amir Gholami, Jiali Yu, Eric Tan, Leyuan Wang, Qijing Huang, Yida Wang, Michael Mahoney, and Kurt Keutzer. 2021. HAWQ-V3: Dyadic neural network quantization. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 11887–11898. PMLR.

Yanqi Zhou, Tao Lei, Hanxiao Liu, Nan Du, Yanping Huang, Vincent Zhao, Andrew M. Dai, Zhifeng Chen, Quoc V. Le, and James Laudon. 2022. Mixture-of-experts with expert choice routing. In

Advances in Neural Information Processing Systems, volume 35, pages 23652–23664.

Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeffrey Dean, Noam Shazeer, and William Fedus. 2022. ST-MoE: Designing stable and transferable sparse expert models. *arXiv preprint arXiv:2202.08906*.

A Mixtral-8x7B Allocation Artifacts

We release per-layer ILP allocations for Mixtral-8x7B-v0.1 (Jiang et al., 2024) at budgets {2.5, 2.75, 3.0, 3.25} under both the baseline and RWS. Full-model PPL evaluation is blocked by a numerical issue: the bf16 stacked-layer forward pass produces activation magnitudes $\sim 3 \times 10^3$ at layer 1, cascading to NaN-valued logits under both flash-attention-2 and SDPA attention implementations. This is a hardware-accumulation limitation common to bf16 inference with MoE outlier activations, not a method or pipeline issue. We provide allocation artifacts for reproducers with FP32-accumulation attention kernels.

B Per-Task Downstream Breakdowns

Table 3: Per-task accuracy on Qwen1.5-MoE-A2.7B.

Task	2.5 b		2.75 b	
	Base.	RWS	Base.	RWS
PIQA	0.7610	0.7590	0.7764	0.7775
HellaSwag	0.6415	0.6575	0.6794	0.6814
ARC-Easy	0.5623	0.5771	0.6014	0.6035
ARC-Challenge	0.3549	0.3704	0.3925	0.3976
WinoGrande	0.5758	0.5887	0.6038	0.6085
LAMBADA-OpenAI	0.5921	0.6205	0.6272	0.6319
LAMBADA-Std	0.6789	0.6380	0.6777	0.6595
Average	0.5952	0.6015	0.6226	0.6257

C RouteQEP Negative-Result Details

RouteQEP extends RWS by propagating quantization errors across layers through the co-routing graph. Let $\delta_{l,e}$ be the local quantization error for expert e at layer l . RouteQEP defines a propagated error:

$$\tilde{\delta}_{l,e} = \delta_{l,e} + \lambda \sum_{e'} \Pr(e' | e, l+1) \tilde{\delta}_{l+1,e'}, \quad (9)$$

where $\Pr(e' | e, l)$ is the empirical co-routing probability—the fraction of tokens routed to e at layer l that are also routed to e' at layer $l+1$.

The recursion gain is approximately $(k \cdot \bar{\pi})^h$, where h is the hop count, k is the top- k parameter,

and $\bar{\pi}$ is the mean routing probability. With $k = 4$ and $\bar{\pi} \approx 0.1$, one hop amplifies by $\sim 2\times$ and two hops by $\sim 4\times$, causing the allocator to concentrate all high-bit slots on the most-amplified blocks.

Results at 2.75 bits on Qwen1.5-MoE:

- **No propagation** (RWS, local): PPL 8.43 (baseline 8.53).
- **1-hop** ($\lambda = 1.0$): PPL 11.62 (+37% vs. baseline).
- **2-hop** ($\lambda = 1.0$): PPL 12.87 (+51% vs. baseline).
- **1-hop normalized** (divide by depth): PPL 10.45 (+23%).

All variants regress PPL. The allocator is responding rationally to an objective that overweights upstream high-traffic experts. We conclude that intra-layer routing information is valuable but cross-layer propagation is not with this formulation.